

IPHY 5800

Advanced Statistics and Research Methods in Integrative Physiology

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	4
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (49.7 / 100)	37.3 / 100

37

CURRENT AVG SCORE
across 6 job roles

76

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces advanced statistical techniques important for analyzing data rising in biomedical research, including physiology. Statistical reasoning will be emphasized through problem solving and applications using statistical software packages.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	49.7	89.7	+40.0	HIGH
ML Engineer	43.6	74.5	+30.9	HIGH
Data Scientist	38.0	77.9	+39.9	HIGH
AI Engineer	37.5	82.4	+44.8	HIGH
Machine Learning Engineer	37.3	63.2	+25.9	MEDIUM
MLOps Engineer	17.9	65.6	+47.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

scale

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems, scale.

2 LOW Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Applied Scientist

49.7 → 89.7 (+40.0 pts)

computational

interdisciplinary

problems

models

technical

learning

systems

industry

ML Engineer

43.6 → 74.5 (+30.9 pts)

learning

machine

systems

models

infrastructure

problems

technical

scale

Data Scientist

38.0 → 77.9 (+39.9 pts)

toolkit

sets

presentation

social

problems

AI Engineer

37.5 → 82.4 (+44.8 pts)

robotics

models

learning

infrastructure

technical

systems

teams

problems

Machine Learning Engineer

37.3 → 63.2 (+25.9 pts)

sets

learning

tools

design

science

models

machine

technical

MLOps Engineer

17.9 → 65.6 (+47.7 pts)

infrastructure

systems

learning

machine

training

tools

models

error

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.